

LC 1541: Minimum Insertions to Balance a Parentheses String

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Core Concept

basically opening bracket → (
and closing bracket →)

Examples

Eg 1:

```
( ( ) ) )  
0 1 2 3 4
```

here 1, 2, 3 are valid
but 0 and 4 makes ()
so it needs another) to make → ())

So o/p = 1

Eg 2:

```
) ) ( ) ) (  
0 1 2 3 4 5
```

→ need one (at -1
2 3 4 balanced
and need)) at 6 and 7

hence o/p = 1 + 2 = 3

Logic & Intuition

very similar to LC 921

- '(' → open++
- ')' → need a second ')' after it

So:

- If open == 0 → ans++ to insert '('
- Else open--
- If next char not ')' → ans++
Else i++ to consume second ')'

C++ Implementation

```
class Solution {
public:
    int minInsertions(string s) {

        int n=s.size();
        int open=0;
        int ans=0;

        for(int i=0;i<n;i++){

            char c=s[i];
            if(c=='('){
                open++;
            }

            else{
                if(open==0){
                    ans++;
                }

                else{
                    open--;
                }

                if(s[i+1]==')'){
                    i++;
                }

                else{
                    ans++;
                }
            }
        }

        ans+=2*open; //every remain ( needs 2 closing )
        return ans;
    }
};
```

